



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

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Azure
Databricks



Absolute Beginner to Specialization in Apache Spark and Azure Databricks



Spark Dataframes - Introduction

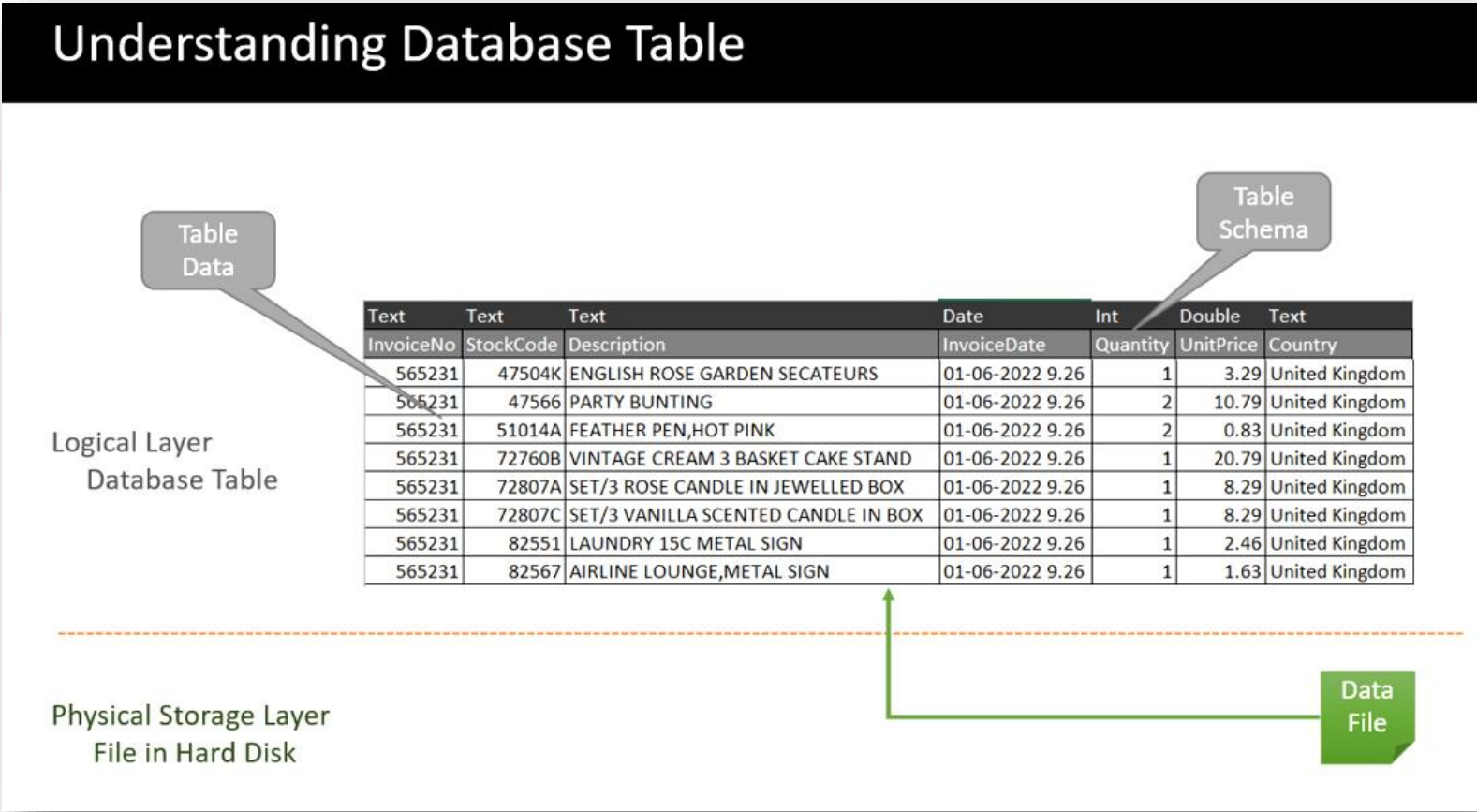
We learned that Spark is a data processing platform.
We also know that Databases are the most popular and most widely used data processing platforms.
They offer two things at a high level.

1. Tables
2. SQL

A database table allows you to load the data in the table. The data in a table is internally stored as a .dbf file, which are stored on the disk. But we don't care about this dbf file and storage layer, we always look at the table, and query the table.

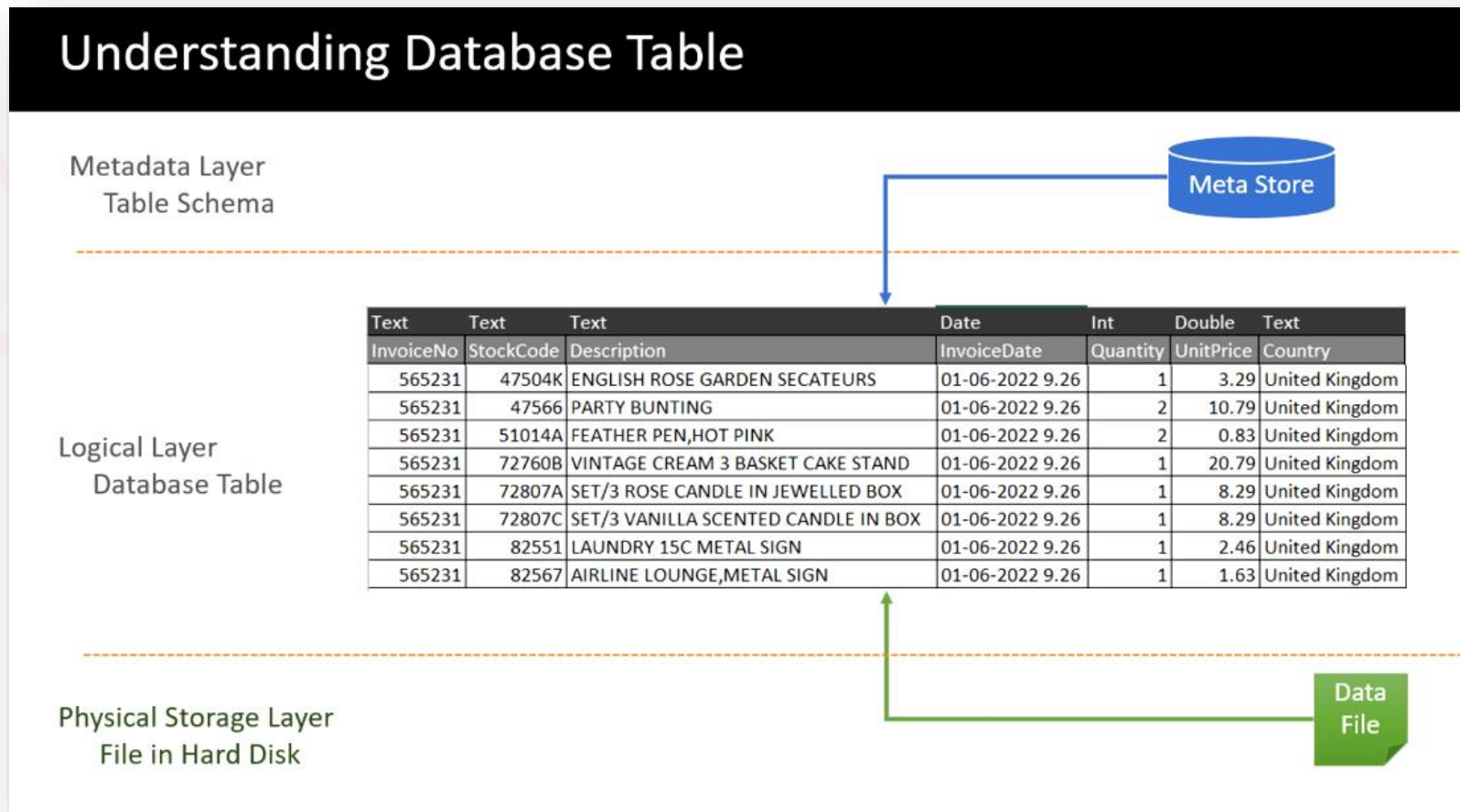
A table contains two things:

- 1. Table Schema – It is a list of column names and data types.
- 2. Table Data – It is the data stored inside the table.



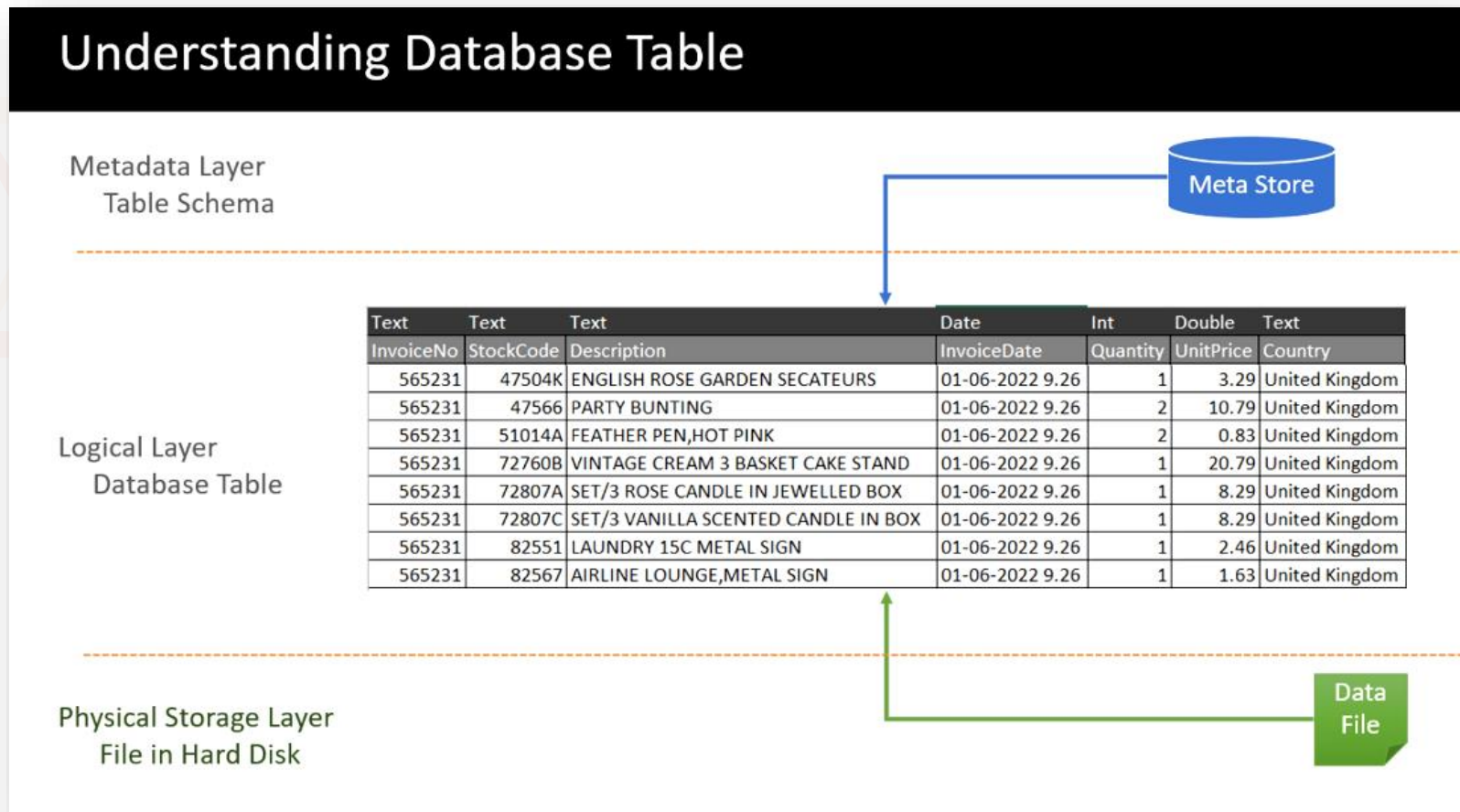
We have three layers to form a table:

1. Storage layer stores the table data in a file
2. Metadata layer stores the table schema and other important information
3. The Logical layer presents you with a database table, and you can execute SQL queries on the logical table.

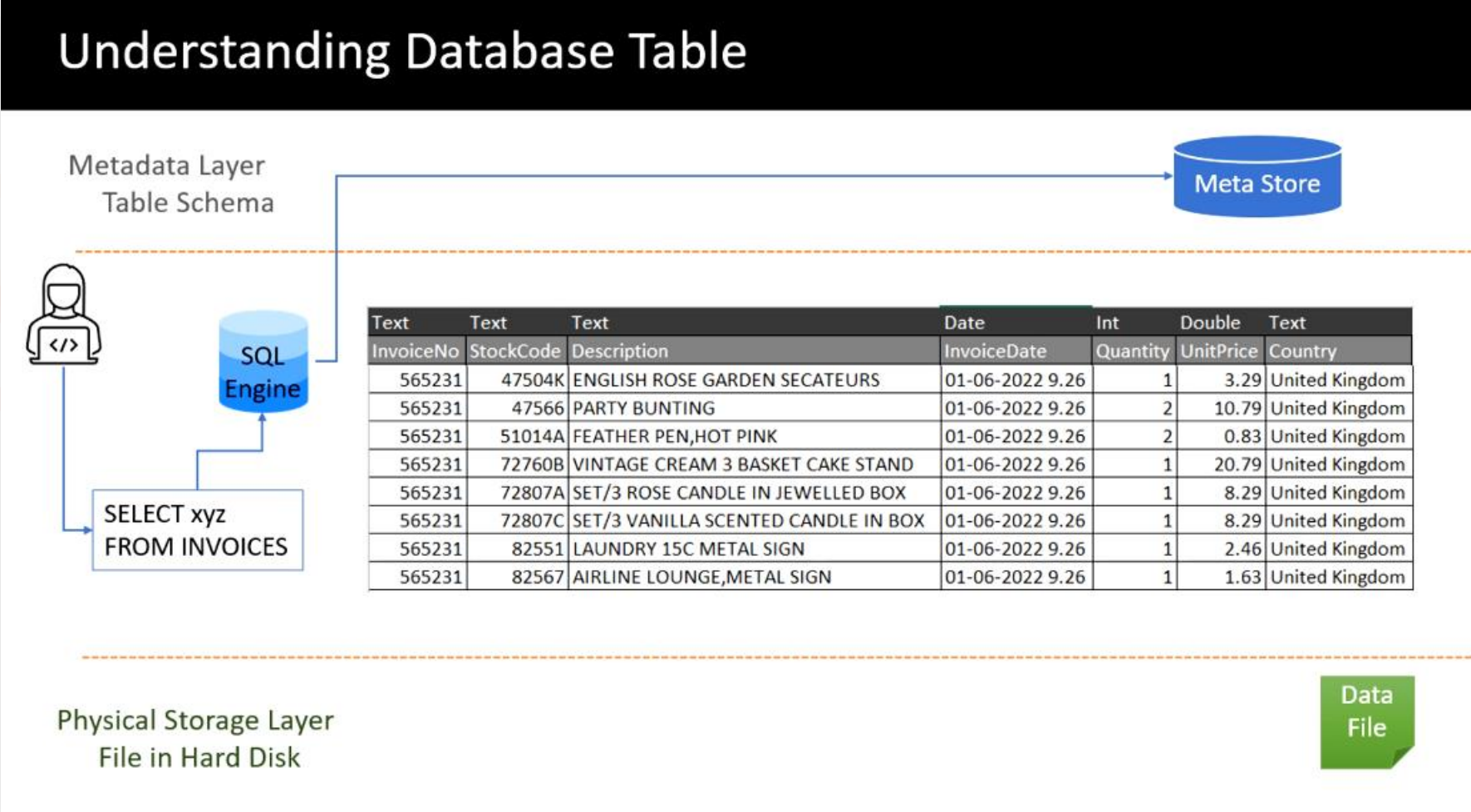


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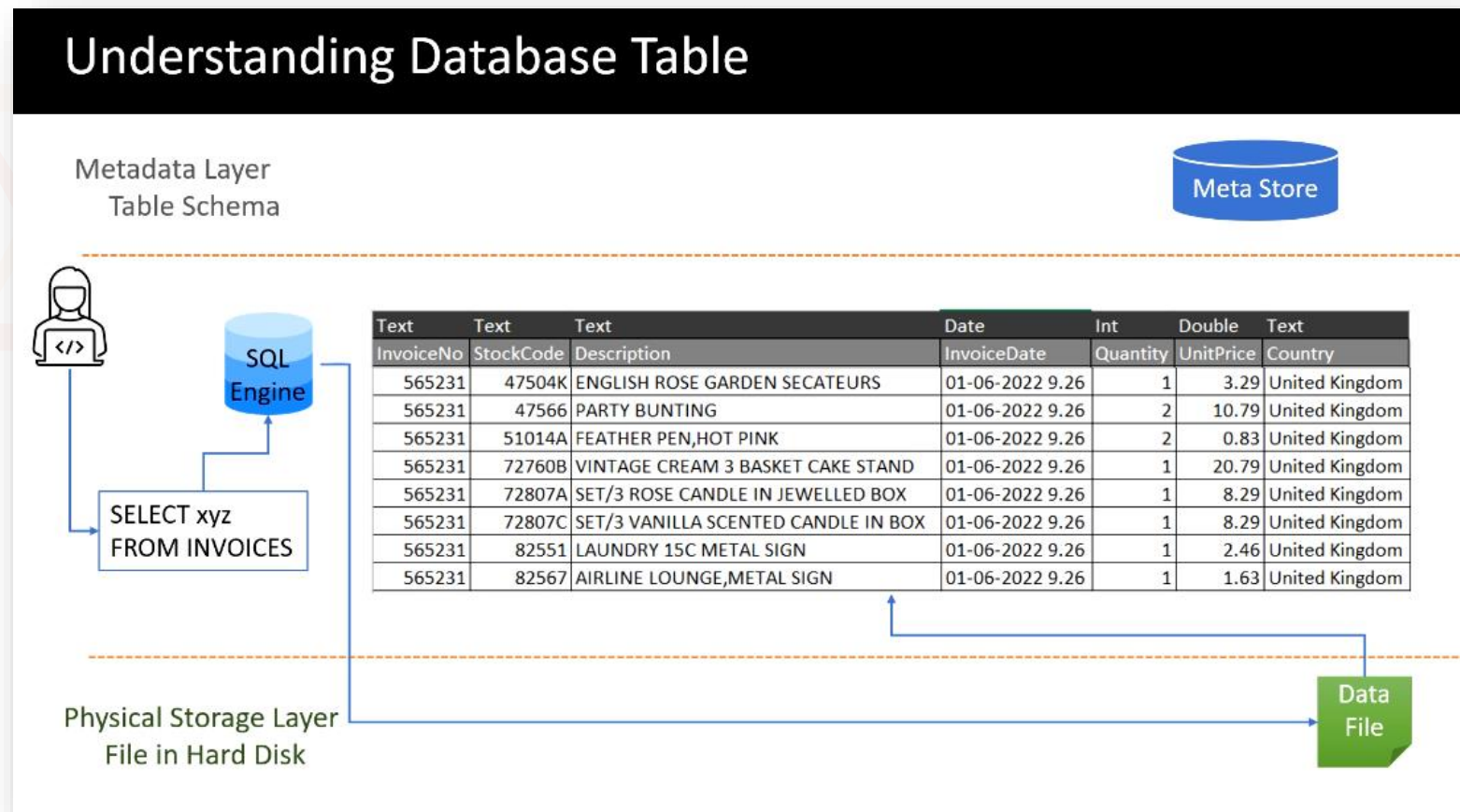


When you submit a SQL query to your database SQL Engine, the database will refer to the metadata store for parsing your SQL queries. And the database will throw a syntax error or an analysis error if you are using a column name in your SQL that does not exist in the metadata store.



The schema is essential for the database table and for the SQL expressions to work correctly. The table data is stored in the dbf files behind the table. So if your SQL query is correct and passes all the schema validation, you will see query results.

The database will read data from the "dbf" file, process it according to your SQL query, and show you the results. And that is all about the data processing in databases.



Apache Spark offers you two ways of data processing:

1. Spark Database and SQL
2. Spark Dataframe and Dataframe API

The first approach is precisely the same as a typical database. You will create table and load data into the table. Spark table data is internally stored in the data files. But these files are not dbf files like in the case of a database.

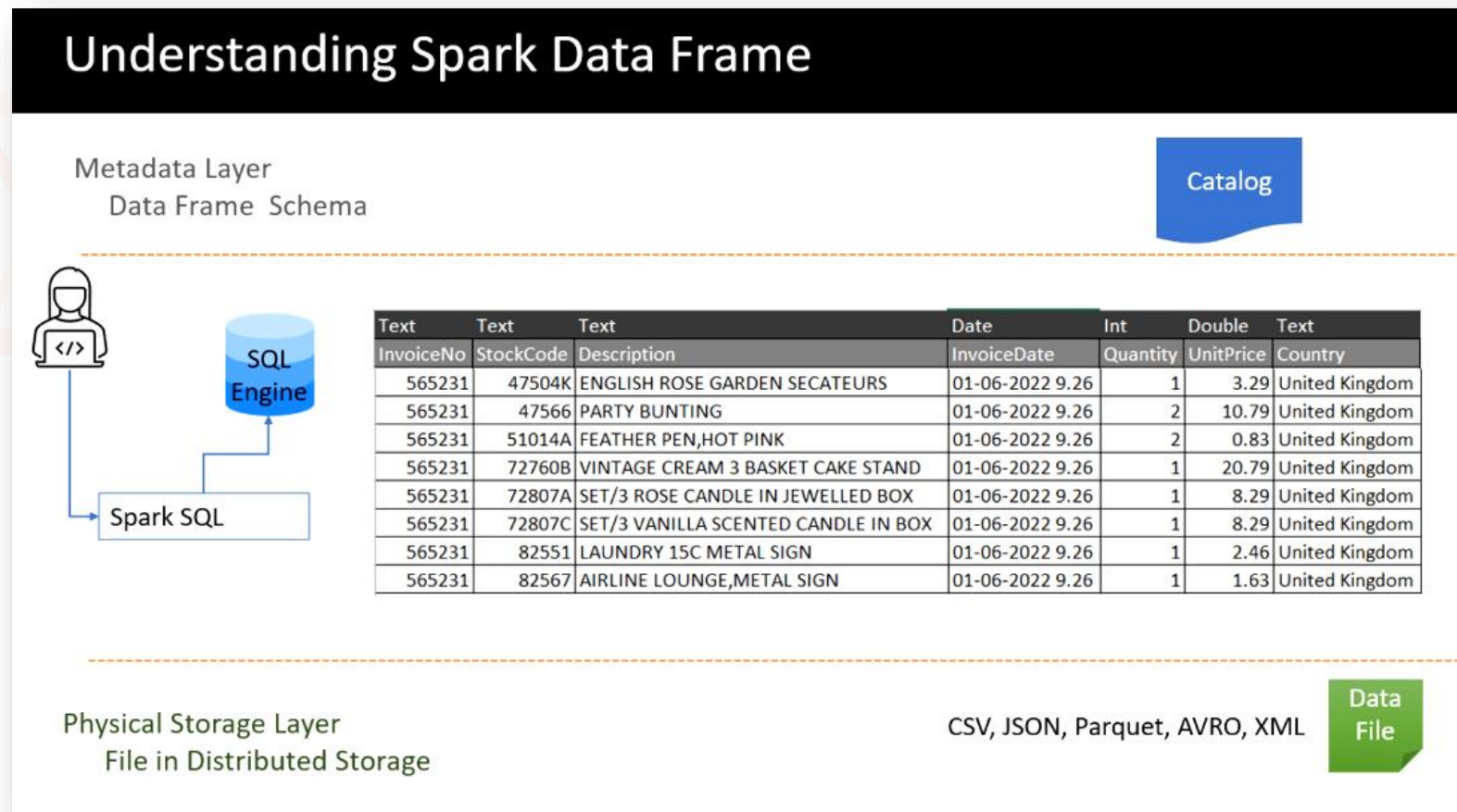
Spark gives you the flexibility to choose the file format and supports many file formats such as CSV, JSON, Parquet, AVRO and XML.

The DBF file format and database engine were designed to support structured data. However, Spark supports structured, semi-structured, and unstructured data.

The spark storage layer also supports distributed storage such as HDFS and Cloud storage such as Amazon S3 and Azure ADLS. So you are not limited to disk storage capacity. You can use distributed storage and store large data files. Spark also has a metadata store for storing table schema information. So that part is similar to the databases.

Then Spark also comes with an SQL query engine and supports standard SQL syntax for processing and querying data from Spark tables.

The second approach to data processing in Apache Spark is **Spark Dataframe and Dataframe API**. Spark Dataframe is structurally the same as the table. However, it does not store any schema information in the metadata store. Instead, we have a runtime metadata catalog to store the Dataframe schema information. It is similar to the metadata store, but Spark will create it at the runtime to store schema information in the catalog.



We have two reasons for storing schema information in the catalog.

1. Spark Dataframe is a runtime object – You can create a Spark Data frame at runtime and keep it in memory until your program terminates. Once your program terminates, your Dataframe is gone. So, it is an in-memory object.
2. Spark Dataframe supports schema-on-read – Dataframe does not have a fixed and predefined schema stored in the metadata store. We load the data into a Dataframe and tell the schema when loading the data. And Spark will read the file, apply the schema at the time of reading, create the Dataframe using the schema and load the data.

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Here is a quick summary discussing the differences between Spark Tables and Dataframes.

Spark Table Vs Data Frame

Spark Table

1. Tables store schema information in metadata store
2. Table and metadata are persistent objects and visible across applications
3. We create tables with a predefined table schema
4. Table supports SQL Expressions and does not support API

Spark Data Frame

1. Data Frame stores schema information in runtime Catalog
2. Data Frame and Catalog are runtime objects and live only during the application runtime. Data Frame is visible to your application only.
3. Data Frame supports schema-on-read
4. Data Frame offers APIs and does not support SQL expressions

Spark Table and a Data Frame are convertible objects



Thank You
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